

1. Read chapter 1 of Russell and Norvig's *Artificial Intelligence: A modern approach* and answer the items below in connection with the *Introduction* lecture.
 - (a) To what extent do the following computer systems embed artificial intelligence:
 - i. Supermarket bar code scanners
 - ii. Internet routing algorithms that respond dynamically to the state of the network.

Answer:

- i. Although bar code scanning is in a sense computer vision, these are not AI systems. The problem of reading a bar code is an extremely limited and artificial form of visual interpretation, and it has been carefully designed to be as simple as possible, given the hardware.
 - ii. This is borderline. There is something to be said for viewing these as intelligent agents working in cyberspace. The task is sophisticated, the information available is partial, the techniques are heuristic (not guaranteed optimal), and the state of the world is dynamic. All of these are characteristic of intelligent activities. On the other hand, the task is very far from those normally carried out in human cognition.
 - (b) Why would evolution tend to result in systems that act rationally? What goals are such systems designed to achieve?

Answer: Evolution tends to perpetuate organisms (and combinations and mutations of organisms) that are successful enough to reproduce. That is, evolution favors organisms that can optimize their performance measure to at least survive to the age of sexual maturity, and then be able to win a mate. Rationality just means optimizing performance measure, so this is in line with evolution.

- - (c) Is AI a science, or is it engineering? Or neither or both? Justify your answer.

Answer: This question is intended to be about the essential nature of the AI problem and what is required to solve it, but could also be interpreted as a sociological question about the current practice of AI research. A science is a field of study that leads to the acquisition of empirical knowledge by the scientific method, which involves falsifiable hypotheses about what is. A pure engineering field can be thought of as taking a fixed base of empirical knowledge and using it to solve problems of interest to society. Of course, engineers do bits of science – e.g., they measure the properties of building materials – and scientists do bits of engineering to create new devices and so on. As described in Section 1.1, the “human” side of AI is clearly an empirical science – called cognitive science these days – because it involves psychological experiments designed out to find out how human cognition actually works. What about the the “rational” side? If we view it as studying the abstract relationship among an arbitrary task environment, a computing device, and the program for that computing device that yields the best performance in the task environment, then the rational side of AI is really mathematics and engineering;

it does not require any empirical knowledge about the actual world – and the actual task environment – that we inhabit; that a given program will do well in a given environment is a theorem. (The same is true of pure decision theory.) In practice, however, we are interested in task environments that do approximate the actual world, so even the rational side of AI involves finding out what the actual world is like. For example, in studying rational agents that communicate, we are interested in task environments that contain humans, so we have to find out what human language is like. In studying perception, we tend to focus on sensors such as cameras that extract useful information from the actual world. (In a world without light, cameras wouldn't be much use.) Moreover, to design vision algorithms that are good at extracting information from camera images, we need to understand the actual world that generates those images. Obtaining the required understanding of scene characteristics, object types, surface markings, and so on is a quite different kind of science from ordinary physics, chemistry, biology, and so on, but it is still science.

In summary, AI is definitely engineering but it would not be especially useful to us if it were not also an empirical science concerned with those aspects of the real world that affect the design of intelligent systems for that world.

- (d) “Surely computers cannot be intelligent – they can do only what their programmers tell them.” Is the latter statement true, and does it imply the former?

Answer: This depends on your definition of “intelligent” and “tell.” In one sense computers only do what the programmers command them to do, but in another sense what the programmers consciously tells the computer to do often has very little to do with what the computer actually does. Anyone who has written a program with an ornery bug knows this, as does anyone who has written a successful machine learning program. So in one sense Samuel “told” the computer “learn to play checkers better than I do, and then play that way,” but in another sense he told the computer “follow this learning algorithm” and it learned to play. So we're left in the situation where you may or may not consider learning to play checkers to be a sign of intelligence (or you may think that learning to play in the right way requires intelligence, but not in this way), and you may think the intelligence resides in the programmer or in the computer.

2. Read chapter 2 of Russell and Norvig's *Artificial Intelligence: A modern approach* and answer the items below in connection with the *Intelligent Agents* lecture.

- (a) For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate:
- i. Every agent is rational in an unobservable environment.
 - ii. A perfectly rational poker-playing agent never loses.

Answer:

- i. False. Some actions are stupid – and the agent may know this if it has a model of the environment – even if one cannot perceive the environment state.
- ii. False. Unless it draws the perfect hand, the agent can always lose if an opponent has better cards. This can happen for game after game. The correct statement is that the agent's expected winnings are nonnegative.

- (b) Give a PEAS description of the task environments below and characterize them in terms of the properties discussed in section 2.3.2 of the textbook
- i. Shopping for used AI books on the Internet
 - ii. Bidding on an item at an auction.

Answer:

- i. Partially observable, deterministic, sequential, static, discrete, single agent. This can be multi-agent and dynamic if we buy books via auction, or dynamic if we purchase on a long enough scale that book offers change.
 - ii. Fully observable, strategic, sequential, static, discrete, multi-agent.
- (c) This exercise explores the differences between agent functions and agent programs.
- i. Are there agent functions that cannot be implemented by any agent program?
 - ii. Suppose we keep the agent program fixed but speed up the machine by a factor of two. Does that change the agent function?

Answer:

- i. Yes; the agent function might specify that the agent print true when the percept is a Turing machine program that halts, and false otherwise. (Note: in dynamic environments, for machines of less than infinite speed, the rational agent function may not be implementable; e.g., the agent function that always plays a winning move, if any, in a game of chess.)
 - ii. It depends on the program and the environment. If the environment is dynamic, speeding up the machine may mean choosing different (perhaps better) actions and/or acting sooner. If the environment is static and the program pays no attention to the passage of elapsed time, the agent function is unchanged.
- (d) Discuss possible agent programs for each of the following stochastic environments:
- i. Murphy's law: twenty-five percent of the time, the Suck action fails to clean the floor if it is dirty and deposits dirt onto the floor if the floor is clean. How is your agent program affected if the dirt sensor gives the wrong answer 10% of the time?
 - ii. Small children: At each time step, each clean square has a 10% chance of becoming dirty. Can you come up with a rational agent design for this case?

Answer:

- i. For a reflex agent, this presents no additional challenge, because the agent will continue to *Suck* as long as the current location remains dirty. For an agent that constructs a sequential plan, every *Suck* action would need to be replaced by "*Suck* until clean." If the dirt sensor can be wrong on each step, then the agent might want to wait for a few steps to get a more reliable measurement before deciding whether to *Suck* or move on to a new square. Obviously, there is a trade-off because waiting too long means that dirt remains on the floor (incurring a penalty), but acting immediately risks either dirtying a clean square or ignoring a dirty square (if the sensor is wrong). A rational agent must also continue touring and checking the squares in

case it missed one on a previous tour (because of bad sensor readings). it is not immediately obvious how the waiting time at each square should change with each new tour. These issues can be clarified by experimentation, which may suggest a general trend that can be verified mathematically. This problem is a partially observable Markov decision process – see Chapter 17. Such problems are hard in general, but some special cases may yield to careful analysis.

- ii. In this case, the agent must keep touring the squares indefinitely. The probability that a square is dirty increases monotonically with the time since it was last cleaned, so the rational strategy is, roughly speaking, to repeatedly execute the shortest possible tour of all squares. (We say “roughly speaking” because there are complications caused by the fact that the shortest tour may visit some squares twice, depending on the geography.) This problem is also a partially observable Markov decision process.
3. Familiarise yourself with the code base for the algorithms presented in the AIMA textbook; the code is available at <https://github.com/aimacode/aima-python>. You have two options to work with aima-python: (1) install Anaconda + Jupyter on your computer – recommended; (2) run aima-python directly online in a Binder Environment – worth trying.
 - (a) Start exploring aima-python with the intro Jupyter notebook (e.g. <https://github.com/aimacode/aima-python/blob/master/intro.ipynb>). If you have a local aima-python (as recommended above) you start with your local version of the intro Jupyter notebook.
 - (b) The intro notebook has a link to another notebook called index (e.g. <https://github.com/aimacode/aima-python/blob/master/index.ipynb>). If you have a local aima-python (as recommended above) you start with your local version of the index Jupyter notebook.

Answer: Follow the instructions in the aima-python code repository.

4. The agents Jupyter notebook (e.g. <https://github.com/aimacode/aima-python/blob/master/agents.ipynb>) has a few examples of agents and environments – i.e. a BlindDog in a park. You will notice that the agents.ipynb imports code from agents.py. This is a common feature in aima-python where Jupyter notebooks use python libraries that actually implement the algorithms from the AIMA textbook.
 - (a) Inspect the code for the abstract Agent and Environment classes which are defined in the `agents.py` library.
 - (b) Starting with a simple BlindDog agent and a Park environment, the agents.ipynb notebook has multiple implementations of BlindDog agents and their Park environments increasing the complexity of the environment. You will learn how complex environments increases the sophistication of the agents e.g. EnergeticBlindDog.
 - (c) The agents.ipynb also shows an example of the Wumpus environment. We will only learn building functionality to reason about individual actions an agent in the Wumpus world can take at any given stage of the agent’s journey in the Wumpus world. A fully

functioning agent for the Wumpus environment is way more complex than building a BlindDog. Why?

Answer: Follow the instructions in the `aima-python` code repository.